UTME & Post-UTME Key Points (SCIENCES)

By: **Peter Eyo**

CHEMISTRY

1. Chlorine is dried by H2SO4
2. Chlorine is obtained by downward delivery (upward displacement of air) because it is denser than air

NOTE:

- Cl_2, CO_2, SO_2 & NO_2 are denser than air
- NH_3 & H_2 are lighter than air

3. Chlorine bleaches by oxidation (permanent bleaching) & SO_2 bleaches by reduction (temporary bleaching). They are used to bleach because they dissolve in water
4. Bleaching by SO_2 is temporary because SO_2 can be oxidized
5. $Pb(NO_3)_2$ =331g/mol
6. A liquid in liquid is emulsion
7. Li & Mg, Be & Al, B & Si, these have a diagonal relationship (To remember, I just say Limg, Beal, BSI (Best Schools International) this is an easy way for me to remember
8. Tea/Coffee stain are removed using Borax Solution
9. Paint is removed using Turpentine
10. Grease is removed using Ammonia or Kerosene
11. Iodine & Naphthalene are removed using Ethanol
12. Carbon-12 is used to check the date of ancient samples
13. Cobalt-60 is used in rapid checking of faults in welding and casting
14. Phosphorous-32 is used to cure or prevent leukemia
15. Iodine-131 is used to prevent Goiter
16. Nitrogen acts as a diluent to slow down combustion and corrosion
17. Polarization is caused by hydrogen in an electrochemical cell and it can be removed by depolarizer such as manganese IV Oxide & Potassium Dichromate
18. Clay is composed of Aluminum, Silicon, Oxygen and Hydrogen
19. The electron contributes least to the mass of an atom or element
20. Chemical properties depend on the atomic number
21. Neon lights are used in aerodrome beacons as they have a high power of fog penetration
22. A mixture of krypton & xenon is used in a photographer's flash tube for taking high speed pictures
23. Argon is used in gas filled electric lamps because it helps to prevent oxidation of the lamp filaments
24. Argo is used for electric arc welding
25. Charcoal is used in decolourization of sugar
26. Lamp black & soot are the same, and are used in the manufacture of lead pencils
27. Graphite is a dry lubricant
28. A non-conductor may be made conductive by coating it with graphite
29. Impurities in a liquid reduce its melting point and increase its boiling point
30. Blood is a mixture and can be separated by centrifugation
31. Pure substances boil and melt at a definite or narrow range of temperature
32. Impure substances melt over a wide range of temperature
33. Melting breaks little or no metallic bonds

34. Boiling breaks all the metallic bonds
35. Petroleum and rubber latex are mixtures
36. Trichloromethane (Chloroform) is used as an organic solvent for Iodine, Alkaloids, Fats & other substances
37. Iodomethane is used as a germicide for skin bacteria
38. Tetrachloromethane is a good solvent used in the dry cleaning industry and in fire extinguishers
39. Methane is used to make Hydrogen & Carbon Black
40. The main constituent of natural gas is Methane
41. Isotones are atoms that have the same neutron number (e.g Chlorine (Cl) & Potassium (K) , Magnesium (Mg) & Sodium (Na))
42. Isobars are elements with the same atomic mass but different atomic number (e.g Potassium, Calcium & Argon = 40, Carbon 14, Nitrogen 14 & Oxygen 14)
43. In Isotopy, the lighter element is more abundant than the heavier element
44. Mass spectrometer is used to determine the relative atomic mass of an element
45. Balancing of equations obeys the law of conservation of mass
46. Real gases deviate at low temperature and high pressure
47. Dispersion, evaporation, diffusion, Brownian motion, Osmosis, Tyndall effect & effusion illustrate the kinetic theory of matter
48. Tyndall effect is the scattering of light by a medium containing small suspended particles
49. The absolute zero temperature is -273°C or 0K
50. H_2SO_4 is used in car batteries
51. HNO_3 is used to make explosives e.g TNT (Trinitrotoluene)
52. Dilute HNO_3 does not react with metals to liberate Hydrogen gas because it is a strong oxidizing agent, it oxidizes the Hydrogen to water
53. Limestone is $CaCO_3$
54. Marble chips is $CaCO_3$
55. Quicklime is CaO
56. Mercury II Nitrate is used as an antiseptic for skin diseases
57. Potassium Bromide is used as a sedative
58. Calcium Sulphate – Plaster of Paris
59. Hydrogen is produced by Bosch process
60. Nitrogen is produced industrially by the Fractional Distillation of liquid Air
61. Chlorine is produced industrially by the electrolysis of brine

A Formula to NOTE:

$$\text{Number of Moles of Water of Crystallization} = \frac{\text{Mass of Water of Cryst. x Molar Mass of Anhydrous Salt}}{18 \quad \text{x Mass of Anhydrous Salt}}$$

18 is the mass of Water (H_2O)
Anhydrous - without water
Hydrated - with water

Mass of Water of crystallization = Mass of Hydrated Salt – Mass of Anhydrous Salt

62. Principal Quantum Number can be used to determine groups, periods, Charge number or Oxidation

number, Valency & Valence electrons

63. Azimuthal Quantum Number can be used to determine the groups, period, Charge, Valency, Valence electrons & Block

64. Quicklime (CaO) is used to dry ammonia (NH_3)

65. A Test for Hydrogen is POP sound

66. In the Fractional Distillation of Liquified Air:

- Oxygen is removed by Alkaline Pyragallol
- Carbon IV Oxide (CO_2) is removed using NaOH (Caustic Soda)
- Water vapor is removed using Calcium Chloride ($CaCl_2$)

67. Oxygen gotten from air is denser than that from the lab because of the presence of Rare Gases

68. Astatine is a Radioactive Halogen (Group & Element)

69. Iodine is Solid

70. Chlorine & Bromine are Liquid

71. Natural Rubber

Monomer is 2-methylbutan-1,3-diene

Polymer is Poly (2-methylbutan-1,3-diene) also known as **Neoprene**

72. Artificial Rubber

Monomer is 2-chlorobutan-1,3-diene

Polymer is Poly (2-chlorobutan-1,3-diene) also known as **Isoprene**

73. Sucrose = Glucose + Fructose (Forward reaction is Hydrolysis, Backward reaction is Condensation)

74. Maltose = Glucose + Glucose

75. Lactose = Glucose + Galactose

76. Nigerian Crude Oil has low Sulphide content, this makes it light

77. The first product of the Fractional Distillation of crude Oil is Natural Gas

78. The Major product of Natural Gas is Methane

79. Pig Iron has a high percentage of impurities

80. The ores of Iron include Haematite, Magnetite, Limonite

81. Ore of Tin is Cassiterite

82. Ore of Lead is Galena

83. Ore of Copper is Copper Pyrite

84. Ore of Zinc is Calamine

85. All Ammonia Salts react with Alkali to liberate Ammonia Gas

86. Nuclear Fission is used in making weapons

87. Nuclear Fusion occurs in Stars

88. $CaCl_2$ is added to NaCl to lower its melting point

89. The raw material used in the smelting of Iron is $CaCO_3$

90. Long chain makes soap Biodegradable

91. The formation of Water Gas is Endothermic

92. The reduction of steam to hydrogen by CO is exothermic

93. Hydrogen is also known as water Former

94. Lead does not react with acids because the lead salts formed are insoluble and form a protective coating over the metal

95. Oxygen is the most abundant element on Earth

96. Industrial prep of Graphite is by Acheson process

97. Wood Charcoal is used to absorb gases (Poisonous Gases), for purification of Noble gases & recovery of

industrial solvents

98. Sugar Charcoal is formed by dehydrating Sugar and is the purest form of Amorphous Carbon

99. Animal/Bone Charcoal has a property of adsorbing coloring matter. It is used in removing the brown color from Crude sugar and decolorizing Petroleum Jelly

100. Catenation is the ability of Carbon to form long chains or rings

101. Carbon combines directly with Sulphur, Hydrogen, Calcium & Aluminium at very high temperatures

102. Solid CO_2 (Dry Ice) is used as a refrigerant

103. Gaseous CO_2 is used to preserve fruits

104. CO_2 is used as a coolant in Nuclear Reactor

105. $BaSO_4$, $PbSO_4$, $HgSO_4$ & $CaSO_4$ are not soluble in water, All other Sulphates are Soluble in water

QUICK NOTES: Note these, they will definitely help you

- Anything Black is Sulphide (H_2S)
- $K_2Cr_2O_7$ – Orange to Green
- $KMnO_4$ – Purple to Colourless
- $CuSo_4$ – White to Blue
- All Ammonium (NH_4) salts react with alkali to liberate Ammonia gas
- Ammonium Ion is NH_4^+
- Ammonia is NH_3
- Brass – Copper (Cu) & Zinc (Zn)
- Bronze – Copper (Cu) & Tin (Sn)

This is how I remember
Bronze has **Z**, the one with **Z** doesn't have Zinc
Brass has **S**, the one with **S** doesn't have Tin which is Sn
They both have copper (they may ask what they have in common)
NOTE that these are **Alloys** (mixture of metals)

ALLOY	ELEMENTS	USES
Brass	Cu & Zn	For making bolts, nuts, moving parts of clocks and watches
Bronze	Cu & Sn	For making coins, medals, sculptures and metal work
Duralumin	Al, Cu, Mg & Mn	Construction of aircraft, ships, cars and moving part of machine
Steel	Fe & C	Construction of bridges, ships, car and machinery
Stainless steel	Fe & Cr	Making cutlery, tools & surgical instruments
Soft Solder		

	Pb & Sn	For welding & Plumbing
Alinco	Fe, Al, Ni, & Co	For making permanent magnets

Table Referenced from **Solution Master Chemistry by <u>Sir Julius Oniya</u>**

106. Terminal Alkyne has triple bond towards the end

NOTE: Only Terminal Alkyne shows positive test with $AgNO_3$/Mirror test

107. Benzene burns with a sooty flame

108. Insoluble salts are prepared by Double decomposition (exchange of radicals) or Direct combination

109. Hg_2Cl_2 is Mercury I Chloride

110. Only Sodium (Na), Calcium (Ca) & Potassium (K) give test to a non-luminous flame, All other ions are tested with NH_3 & NaOH

111. Effects of radiation include cell damage, genetic abnormality, & Cancer

112. The purest form of Iron (Fe) is wrought iron

113. Pig Iron has a high percentage of impurities

114. Polysaccharides include glycogen, starch, cotton, cellulose etc.

115. The purest form of natural water is rain water

116. The purest form of treated water is distilled water

117. CO_2, SO_2 & NO_2 form acid rain

118. PURIFICATION OF WATER

- Aeration: removal of dissolved gases/gaseous pollutants
- Addition of Alum [$KAl(SO_4)_3$ or $NaAl(SO_4)_3$] : Alum coagulates fine particles (Clumping together of particles or removal of small solid particles)

NOTE: The small particles come together & become bigger, forming a large mass

- Sedimentation: settling down of the clumped particles
- Filtration: to separate the filtrate from the residue
- Disinfection:

- Chlorine: To kill germs
- Fluorine: To prevent tooth decay
- Iodine: To prevent Goiter
- $KMnO_4$: To kill Bacteria
- $CuSO_4$: To prevent the growth of Algae (mostly in pools)
- $Ca(OH)_2$ (slaked Lime): To control the pH of water or improve the taste

119. Oxygen is prepared industrially by the fractional distillation of liquefied air

120. Constituents of Air

- N_2
- O_2: removed using Alkaline Pyrogallol
- CO_2: removed using NaOH (caustic soda)
- Water Vapour: removed using $CaCl_2$
- Rare gases

Note: Oxygen gotten from air is denser than that from the lab because of the presence of rare gases which cannot be removed because they do not react

121. The most electronegative element if Fluorine

122. The hydrolysis of NH_4Cl salt will give an acidic solution
123. Pollutants that are likely to be present in an industrial environment are H_2S, SO_2 & oxides of Nitrogen
124. During electrolysis:

- **At the Cathode (negative electrode)**: The product liberated is either a metal or hydrogen
- **At the Anode (positive electrode)**: The product is oxygen, a non-metal or the anode may dissolve in solution

125. Redox reaction takes place in Daniel cell
126. An effect of thermal pollution on water bodies is the reduction in the level of oxygen
127. The arrangement of particles in crystal lattices can be studies using x-ray
128. Sugar is separated from its impurities by crystallization
129. Elements in the same period in the periodic table have the same number of shells
130. Filtration is used to separate an insoluble solid from a liquid
131. A likely consequence of using fuel with high sulphur content is acid rain

BIOLOGY

1. Duck Billed Platypus:
- An egg laying mammal (Monotreme)
- Do not have Mammary Glands
- Their sweat Glands secrete milk
- They have Beaks just like Birds
2. Undulation is a wave like movement found in Earthworms & Snakes
3. Movement in White Blood Cell (WBC) is Amoebic
4. Respiration is an oxidation Process
5. Mitochondria is the power house of the cell
6. Respiration takes place in the part of the lungs called Alveoli
7. Erythrocytes (Red Blood Cells) transport oxygen
8. Fermentation respiration is a type of Anaerobic respiration (Oxygen Isn't necessary for break down) in which sugar is converted into ethanol (alcohol) & Carbon IV Oxide e.g Respiration in Fungi
9. Lactic respiration is a type of Anaerobic respiration in which simple sugar is converted into Lactic acid
10. The Glycolytic pathway of Anaerobic respiration is located in the Cytoplasm
11. Gaseous Exchange
- Earthworm & Amoeba use simple diffusion
- Bird use Air Sac
- Man uses Lungs
- Insects use Trachea
12. Lungs in mammals are enclosed by Pleura membrane
13. The essence of tissue respiration is to distribute oxygen to the body cells
14. Oxygen which serves as a by-product of photosynthesis is derived from the photolysis of water
15. Etiolated plants are influenced by light
16. Parasitic plants include Mistletoe, Dodder, cassytha, Striga etc.
17. Ectoparasites include Fleas, Tick, Aphids
18. Endoparasites include Plasmodium, Tapeworm, Hookworm, Trypanosome, Flukes
19. Carnivorous plants include Pitcher plant, Bladderwort, Sundew & Venus fly trap
20. Irritability is the ability of living organisms to detect and respond to external & internal changes (Stimuli)
21. The response of plants to suppor5t (touch) is called Thigmotropism or Haptotropism
22. The response of plants to salt is Halotropism
23. The response of plants to light is Phototropism
24. The shoot of plants show positive phototropism (grow towards light)
25. The root of plants show negative phototropism (grow away from light)
26. Apical meristem brings about Primary growth in Plants
27. Cambium brings about Secondary growth in plants
28. Growth takes place when the rate of Anabolism (build up) exceeds the rate of catabolism (break down)
29. Excretory products in Plants include Alkaloids, Gum, Resin, Mucilage, Tannin & Oxygen
30. Excretory products in Animals include Sweat, Water vapor, Uric acid (In Insects), Urea (In Mammals), Urine

NOTE: Faeces is not an excretory product but an "Egestion"

Egestion is the removal of the undigested part of the food in the body

31. Excretory organs
- Earthworm – Nephridium
- Tapeworm – Flame cell
- Insects – Malpigian Tubules
- Crabs – Green glands
- Arachnids – Lung book
- Amoeba, Paramecium – Contractile vacuoles (gets rid of excess water)
- Mammals – Kidneys
- Leaves – Stomata

32. Binary fission is the type of reproduction common to Amoeba & Paramecium
33. Parthenogenesis is the production of offspring's or young ones without fertilization e.g Insects (Aphids)
34. Parthenocarpy is the formation of fruits from unfertilized ovary e.g In Pineapple
35. Paramecium uses trichocyte for offense & defense
36. Hydra uses Nematocyst or Cnidoblast to paralyze its prey
37. The Toad uses poison gland to scare or frighten its enemies
38. The Lizard lowers its gular fold to scare or frighten its predators
39. Chameleon uses a photosensitive receptor called Chromatophore to change its color & hide from predators
40. Fish use scales to protect itself
41. In Cactus Opuntia, a desert plant (Xerophyte) leaves are reduced to spines to conserve water
42. Xerophytes are plants that grow in places with little or no water
43. Basking in Lizard is to activate or Raise their body temperature (When Lizards expose their body to the sun)
44. Feathers in Birds are also used for Insulation
45. Feathers in Peacock is to court females (attract them)
46. Rainbow head or Red head of the male agama lizard is to secure its mate
47. Agama lizards show territorial behavior
48. Aestivation is a condition in which an organism enters into a state of dormancy during the dry/hot season (deep sleep)

This is observed in Snails, Frogs, and Lung fish (A Fish capable of Aestivation)

49. Blood is a liquid tissue
50. Bones are tissues
51. The Stapedus (smallest muscle) is found in the middle ear and attached to the Stape (smallest bone)
52. The largest muscle is the glutea, found in the buttocks
53. The skin, nail & hair are part of the integumentary system
54. Volvox, Eudorina & Pandorina exist in Colonies
55. Guard cells which form the opening into Stomata have bean shaped structure
56. Mitochondria have rod shape
57. The Protoplasm mean living cell or living material in the cell and it consists of the Nucleus & Cytoplasm
58. The Nucleus houses the basic unit of inheritance (heredity)

59. In the Brain, the respiratory center is located in the Medulla Oblongata
60. In the Lungs, the respiratory zone is located in the Alveoli
61. The rough Endoplasmic Reticulum contains Ribosome on its surface and is involved in the production of protein
62. Ribosome is involved in protein synthesis
63. The Cell vacuole is enclosed by a membrane called **Tonoplast**
64. Chloroplast is an important plant plastid and is enclosed by a membrane called **Thylakoid membrane**
65. Amoeba, Paramecium & Euglena do not have respiratory structures because the ratio of their surface area to that of the volume is great. They depend on simple diffusion through the cytoplasm
66. The principal mode of gaseous exchange in Earthworm is diffusion
67. The exchange of material between the mother blood and foetus through the placenta involves diffusion
68. Osmosis is responsible for the absorption of water by the roots of plants
69. The rate of Transpiration in plants can be measured and determined by a device called Potometer
70. Small pox, Chicken pox, Bird flu, Lassa fever, Ebola, Hepatitis, Measles, & Corona are all caused by Virus

NOTE: Virus can ONLY be cured by vaccines, not by Antibiotics

71. Bacteriophage is a Virus that infects a bacteria, It contains only DNA
72. Ruminant animals do not have Cellulase that helps them digest Cellulose in Plants, it is the Bacteria that helps them
73. Syphilis (Treponema pallidum), Tuberculosis (Mycobacterium tuberculosis), Typhoid fever (Salmonella typhi), Gonorrhea (Gonococcus), Ulcer, Cholera (Vibrio cholera) & Leprosy (Tetanus) are caused by Bacteria
74. Pasteurization is the process in which milk is heated at 78°C for 30 minutes to kill the Bacteria
75. Preservation of food using refrigerator is to render the bacteria in the food inactive at low temperatures
76. Mycorrhizae is an association between Fungi and the root of higher plants
77. Xylem conduct water & mineral salts from the roots of the plant to the other parts of the plant
78. The Mangrove plant uses Pneumatophore (Breathing roots) for gaseouws exchange
79. Radial symmetry: is the body division in which an organism is divided into 2 equal parts or identical parts along any plane or transverse plane

e.g Hydra, Starfish, Obelia, Jellyfish

80. Bilateral symmetry: is the type of body division in which an organism is divided into 2 equal parts along a single plane which is the longitudinal plane

e.g Insects, All vertebrates, All Worms

81. The fertilization of Tapeworm takes place in the proglottides
82. The fertilization of Earthworm takes places in the clitellum in a structure called cocoon
83. Mollusks have muscular foot which is the central part modified or adapted to locomotion (Creeping)
84. Chitin is strengthened by protein
85. Ecdysis is the shedding off of exoskeleton for growth to take place

e.g In Snakes, Insects etc.

86. Complete metamorphosis (have pupa stage) is observed in Butterfly, Housefly, & Mosquito
87. Incomplete metamorphosis (have no pupa stage) is observed in Cockroach, Grasshopper, & Termite
88. Early stage of Butterfly (caterpillar) is destructive because it feeds on leaves. The adult Butterfly helps in Pollination
89. Sperm are produced in the Seminiferous tubules
90. Sperm is stored temporarily in the Epididymis
91. Fertilization takes place in the Oviduct

92. If the egg is fertilized, it becomes implanted in the walls of the Uterus
93. Sperm & Urine both pass through the Urethra, ONLY Urine passes through the Ureter
94. A thin barrier between the maternal & foetal blood supplies, known as **Trophoblast**, allows a selective exchange of materials between them
95. Oxygen & nutrients diffuse into the foetal blood from the maternal blood to be used for the growth and development of the foetus
96. Ultrafiltration takes place in the bowman's capsule
97. A long narrow tube called Ureter connects the kidney to the urinary bladder where Urine is stored temporarily
98. Hydrometer is used to measure specific gravity or relative density
99. Hygrometer is used to measure water vapour in the air
100. Acoelomates include Coelenterates (Hydra) & Platyhelminthes/Flatworms (Tapeworm & Flukes)
101. Coelomates include Annelids (Earthworm), Arthropods (Insects), Molluscs (Snail), & Mammals
102. Pseudo-Coelomates include nematodes (Hookworm)
103. Coelenterates are diploblastic
104. Triploblastic organisms include worms (Platyhelminthes, Nematoda, Annelida), Vertebrates, Molluscs & Arthropods
105. Blood Fluke causes Bilharziasis/Schistosomiasis
106. Filarial worms cause Elephantiasis/Filariasis and the vector is the Culex mosquito
107. Ochocerca (Filarial worm) causes Onchocerciasis/River blindness and the vector is Blackfly
108. Plasmodium causes malaria and the vector is the female Anopheles mosquito
109. The largest animal kingdom phylum is Arthropoda followed by Mollusca
110. The pupa stage of a butterfly is called Chrysalis
111. Amphibians are the first terrestrial vertebrates
112. Reptiles are the first terrestrial vertebrates which are fully adapted
113. The only vertebrates that undergo metamorphosis are the Toads, Tadpole & Frog
114. Crocodile has 4 heart chambers
115. All Reptiles have homodont (same size of teeth) dentition except crocodile
116. The feather used for flight is the Quill
117. Only mammals have sebaceous gland, pulmonary circulation, mammary glands & hair
118. Crossing over occurs at Prophase 1 of Meiosis 1
119. Blood group A, B & AB have Antigens\
120. Blood group A, B & O have antibodies
121. Genetic material/information is located on the locus of chromosomes
122. Ability to grow facial hair is determined by the X Chromosome
123. Ability to grow long hair is determined by the Y chromosome
124. Fertilization takes place at the lower part of the Oviduct/Fallopian tube
125. Zygote is diploid (2n)
126. Back cross: Cross between offspring & the first filial generation, with recessive
127. Test cross: Cross between 2 first filial generations with recessive or dominant parents
128. Cross breeding: Cross between 2 parents with contrasting genes/characters\
129. Monohybrid cross: 1 trait or gene in the cross
130. Dihybrid cross: 2 traits or genes in the cross
131. Sex linked characters are caused by recessive genes on the X chromosome
132. Sex linked diseases include:

- Sickle Cell: Abnormal shape of Red Blood Cell (RBC)
- Haemophilia: Deficiency of Blood clotting factors e.g Fibrinogen

NOTE: Haemophilia affects the sons of the family, while the females are carriers

- Colour Blindness: The infected person has problems with Green & Red colour and it affects the males more than females
- Baldness
- Albinism

133. Night blindness is caused by deficiency of Vitamin A
134. Melanin is produced by the malpighian layer, it absorbs UV light
135. Organic mutation was discovered by Hugo Devries
136. Charles Darwin & Wallace propounded the theory of Natural selection
137. Development of humans follows a trend

FARAM (Fishes→Amphibians→Reptile→Aves→Mammals)

There's a question in Biology asking where humans most recently evolved from

138. Vestigial parts/organs (no longer have use in the body)

- Tail in Man
- Appendix in Man
- Toes in Horse
- Pineal
- Limbs in Snake

139. Amoeba moves by means of Pseudopodia also known as false foot
140. Components of blood are Red Blood Cell (RBC), White Blood Cell (WBC), Plasma & Platelets
141. Commensalism is a relationship found between Shark & Remora fish. Only the Remora gains, the Shark isn't affected
142. The relationship between tapeworm & Intestine of man is parasitic
143. 55% of the blood is made up of plasma
144. 92% of the total plasma is made up of water
145. Birds communicate by contact notes
146. The mammalian heart is divided into two by a muscle known as septum
147. The right auricle and ventricle are separated by the tricuspid valve
148. The left auricle and ventricle are separated by the bicuspid valve
149. The semi lunar valve is found in the aorta & pulmonary artery
150. ANIMAL HORMONES

Pituitary Gland (Base of the mid-brain)

- Prolactin – Stimulates & controls milk production
- Oxytocin – Controls contraction & the flow of milk
- ADH (Vasopressin) – Stimulates kidney tubules to reabsorb water from the glomerular filtrate
- Somatotropin (Pituitarin) – Promotes growth of bones & muscles

NOTE: The pituitary gland controls the secretions of other endocrine glands hence why it is called the master gland

Thyroid Gland

- Thyroxin
- Regulates the rate of metabolism, especially respiration
- Stimulates mental & physical growth and development in young animals
- Controls metamorphosis in tadpoles

Parathyroid Gland

- Parathormone
- Controls the level of calcium content in the blood

Pancreas

- Insulin
- Maintains blood sugar level
- It converts glucose to glycogen by stimulating the liver
- Glucagon
- Increases blood sugar by stimulating the liver & muscle cells to reconvert glycogen to glucose for oxidation to release energy

Adrenal Gland

- Adrenalin
- Aids dilation of the pupils
- Increases heartbeat & respiration
- Increases sugar content of the blood
- Responsible for shivering during cold

Stomach

- Gastrin of the stomach
- Activates gastric gland to produce gastric juices

NOTE: The Pancreatic Juices are Trypsin, Amylase & Lipase

Testes

- Testosterone
- Stimulates the appearance of secondary sexual characters in males at puberty (pubic hairs, beard, thick muscles & deep voice)
- Stimulates the production of spermatozoa by the testes
- Stimulates the development of the male reproductive organ (The Penis)

Ovaries

- Oestrogen
- Stimulates the development of female secondary characteristics at puberty (pubic hairs, smooth skin, uterus growth & breast development)
- Progesterone
- Prepares the uterus for the attachment of the embryo
- It maintains the foetus during the development in the uterus, hence it is called the **pregnancy hormone**

151. Divergent evolution: The pattern of evolution in which 2 related organisms become dissimilar over time because they have adapted to different environments e.g Wings in Eagle & Penguin
152. Convergent evolution: The pattern of evolution in which 2 unrelated organisms become similar because they have adapted to the same environment e.g Wings in Bat (Mammal) & Bird (Aves), Streamlined body in Whale (Mammal) & Fish (Pisces)
153. Co-evolution: Two organisms evolve together e.g Flower & Butterfly

PHYSICS

1. Scalar Quantities (have only Magnitude, no direction):
- Length/distance
- Mass
- Volume
- Density
- Time
- Speed
- Temperature
- Energy
- Electric potential difference
- Electric potential
- Magnetic flux
- Magnetic flux density
- Area
- Work
- Electric charge
- Altitude
- Kinetic energy
- Potential energy
- Electric capacitance
- Pressure
- Power

2. Vector Quantities (have Magnitude & Direction):
- Displacement
- Velocity
- Acceleration
- Weight
- Force
- Momentum
- Electric field
- Magnetic field
- Gravitational field
- Magnetic field intensity\
- Electric field intensity
- Acceleration due to gravity
- Moment
- Torque
- Impulse

3. The Diode valve is a non-ohmic conductor
4. The Angle of dip increases from 0° at the magnetic equator to 90° at the magnetic poles
5. The eye has a crystalline biconvex (Converging) lens

6. PARFID: For similarities between the Camera & the Eye

- Pupil & Aperture
- Retina & Film
- Iris & Diaphragm

NOTE: When an **Astronomical Telescope** is at **normal adjustment**, the final image is **at infinity**

7. A Simple microscope is a convex lens used to produce a magnified image of small objects

NOTE: When a convex/converging lens is used as a simple microscope, the object is placed between the **Principal focus** and the **Optical center** of the lens

A **magnified**, **virtual** & **erect image** is formed

8. The compound microscope is made up of a combination of two convex lenses of short focal length

The lens nearer the object is the **objective lens**

The lens through which the final image is seen is the **eyepiece**

The **eyepiece** has a **larger focal length** than the **objective lens**

The eyepiece acts as a **magnifier**

9. Diamonds achieve their brilliance from a combination of Dispersion and Total Internal Reflection
10. The mass of a substance is constant but the weight may vary on different points of the earth

Due to the fact that the acceleration due to gravity is slightly less in higher areas (mountains)

Weight=mg

11. One special advantage of alcohol over mercury as a thermometric fluid is it's low freezing point
12. Specific latent heat of vaporization of a substance is always greater than its specific heat of fusion
13. A vapour is said to be saturated when a dynamic equilibrium exists between liquid molecules and the vapour molecules
14. For semi-conductors, as temperature increases, conductivity increases due to the hole-electron pair generation
15. Semi-conductors belong to group 4 (They have 4 valence electrons)
16. For metallic conductors, increase in temperature results in a decrease in metallic conductivity
17. The bond in semi-conductors is covalent
18. Insulators have very high resistivity e.g Quartz, Polyethylene & Diamond. They have no free electrons to participate in conduction
19. A plumbline is used to determine the center of gravity of an irregular object
20. Application of total internal reflection is seen in the field view of a fish under water & the formation of mirage
21. Monochromatic light is light of 1 wavelength
22. For Total Internal Reflection to occur

- Light must be travelling from a more dense to a less dense medium
- The critical angle must be exceeded

23. The critical angle is the angle of incidence when the angle of refraction is 90°
24. In Total Internal Reflection, all the rays are reflected
25. The battery connection that permits current to flow across a P-N junction is called forward bias
26. When the base current of a common emitter transistor is kept at zero, it operates in the cut-off region
27. Diode is an electrical device that allows the flow of current in one direction

28. Zener Diode is used for voltage regulation & stabilization
29. Charges will concentrate at the sharpest point
30. After reflection from a concave mirror (converging), rays of light from the sun converge at the focus
31. In the construction of induction coil, the capacitor is used to prevent sparking between the gaps
32. 1 Armstrong = 1 Å = 10^{-10}m
33. Weak radiation of short wavelength causes electron emission immediately
34. Fluorescent lamps are more efficient than filament lamps
35. The acceleration due to gravity:
- Is minimum at the equator
- Increases with altitude reaching the maximum value at the poles of the earth

36. Wave Nature of Light
- Interference
- Diffraction

37. Particle Nature of Light
- Photoelectric effect
- Emission & absorption of light
- Compton scattering
- Thermionic emission

38. In solid friction, the limiting frictional force is:
- Independent of the surface area in contact for a given object
- Roughly proportional to the normal reaction of the surface

39. To get the maximum range, a projectile should ideally be projected at an angle 45°
40. An object will float in a fluid if the average density of the object is less than the density of the fluid. An object will sink in a fluid if the average density of the object is greater than the density of the fluid
41. If an unknown charge is brought near a charged electroscope and the leaf opens more, the unknown charge is similar to the charge on the electroscope
42. Negative charge is the charge on an ebonite rod rubbed with fur or on a rubbed polythene strip
43. Positive charge is the charge on a glass rod rubbed with silk or on a rubbed cellulose acetate strip
44. The Diode is an electric device that allows the flow of current in one direction
45. A Diode can act as a rectifier when it is forward-bias
46. A Zener diode is used for voltage regulation & stabilization
47. Transistor is a three terminal electrical device which helps in the amplification of current
48. The 3 terminals of a transistor are Base, Emitter & Collector
49. A transistor is used in the rectification of signals because it controls the flow of current
50. The battery connection that permits current to flow across a P-N junction is called forward-bias
51. Mechanical waves require a medium for their propagation
52. Electromagnetic waves do not require a medium for their propagation
53. A stationary wave is obtained when two progressive waves of equal amplitude & frequency are travelling in opposite directions and combine together
54. During refraction, the speed of the wave remains the same
55. Beats are used to determine the frequency of a tuning fork and to tune an instrument e.g A Piano
56. The process of producing transverse vibrations which are only in one plane is known as Polarization
57. Mirrors in a Kaleidoscope are inclined at an angle of 60°
58. The number of images formed in a Kaleidoscope is 5 [number of Images (n)= $\frac{360}{\theta}-1=\frac{360}{60}-1=$ 5 images]

59. The periscope has 2 parallel mirrors inclined at 45°

60. Convex mirror & Concave lens → Diverging

- The images formed are always virtual, erect & diminished

61. Concave mirror & Convex lens → Converging

- The images formed depend on the position of the object

62. The Ionic theory was developed by Arrhenius (Swedish scientist)

63. A voltameter is the vessel containing the electrolyte & the electrode

64. During electrolysis:

- **At the Cathode (negative electrode)**: The product liberated is either a metal or hydrogen
- **At the Anode (positive electrode)**: The product is oxygen, a non-metal or the anode may dissolve in solution

65. In the electrolysis of acidulated water using platinum electrode

- **oxygen** is formed at the **anode**
- **Hydrogen** is formed at the **cathode**

66. The amount of substance liberated at either electrode depends on:

- The mass of substance liberated
- The current passed through the electrolyte
- The time for which the current is passed

67. APPLICATIONS OF ELECTROLYSIS

- Electroplating
  - The material to be plated is placed as the cathode
  - The coating material is made the anode of the voltameter

NOTE: The need for electroplating a material is to improve appearance and prevent corrosion

- The purification of metals (eg. Copper)
  - The impure metal is made the anode
  - The pure metal is made the cathode

NOTE: Pure Copper is used in the manufacture of connecting wires & electric cables because of its low resistance

- Preparation of metals from their compounds
  - Electrolytic processes can be used to prepare metals such as Aluminium, Sodium & Potassium from their molten chloride or hydroxide solutions

68. Experiments with discharge tube show that gases conduct electricity under low pressure & high voltage/P.d (greater than 100V)

69. Nature & properties of cathode rays:

- They consist of streams of fast moving particles of negative electricity called electrons
- They cause materials (Glass or Zinc) to fluoresce
- They travel in straight lines
- They are deflected by electric & magnetic fields
- They can ionize gas
- They can affect photographic plates
- They can produce X-rays from high density metals when they are suddenly stopped by such means
- They are highly penetrating (can penetrate aluminium plate & even steel plate)

70. Fluorescent tubes produce light by excitation of gas molecules

71. Cathode rays are produced in a gas discharge tube under very low pressure & high voltages

72. Interference & Diffraction show wave behavior
73. Emission & Absorption of photons demonstrate particle behavior
74. Electrons are emitted from a metal provided that the wavelength of the radiation is less than a certain value, depending on the kind of metal
75. The type of cell used in most photographic light meters produces its own voltage and needs no battery or power supply, Such cells are known as **photovoltaic cells**
76. Photoelectric effect: refers to the emission or ejection of electrons from the surface of a metal in response to incident light

NOTE: This takes place because of the energy of incident photons of light have energy more than the work potential of the metal surface, ejecting electrons with positive K.E

77. Wave Nature of Light:
- Interference
- Diffraction

78. Particle Nature of Light:
- Photoelectric effect
- Emission & Absorption of light
- Compton scattering
- Thermionic emission

79. Electromagnetic radiation behaves both as a waves & particle
80. Electromagnetic waves travel at the speed of light, Mechanical waves travel at a velocity less than the speed of light
81. Electromagnetic waves exhibit reflection, refraction, interference & diffraction
82. Gamma rays are used in the medical field to sterilize medical supplies like surgical equipment, drug, food, bandages etc.
83. Electromagnetic waves are always transverse
84. Mechanical waves can be transverse or longitudinal
85. Gravitational forces are always those of attractions
86. Planets → MVEMJSUNP (I say em vem J sun P to remember)→ Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto
87. Terrestrial planets → MVEM (They resemble the earth and are primarily composed of rocks & metals)
88. Gas giants → JSUN
89. The two largest planets Jupiter & Saturn are composed mainly of Hydrogen & Helium
90. Nuclear fusion reaction produces the solar energy of the Sun
91. Venus spins in an opposite direction
92. The Earth's moon is the largest natural satellite in the solar system
93. The first artificial satellite Sputnik 1 was launched in 1957 by the then Soviet Union
94. Escape velocity is the minimum velocity required for an object (e.g Satellite or rocket) to just escape or leave the gravitational influence or field of an astronomical body (e.g Earth) permanently
95. To escape the gravitational influence of the earth, the satellite must be projected with a velocity greater than $11Kms^{-1}$
96. Acceleration due to gravity on the moon is $\frac{1}{6}$th that on the earth
97. All metals obey Ohm's law
98. Magnetic substances include Iron, Steel, Cobalt, Nickel & Manganese
99. Non-Magnetic substances include Copper, Brass, Wood, Glass

100. Magnets made from soft iron lose their magnetic properties easily (temporary magnet)
101. Magnets made from steel do not lose their magnetism easily (permanent magnet)
102. Primary cells are not rechargeable e.g LSD → Leclanché cell, Simple cell, Daniel cell
103. Secondary cells are rechargeable e.g Lead-acid accumulator, Alkaline/Nickel-Iron (NIFE) accumulator
104. A Galvanometer is used for detecting and measuring very small currents
105. A Galvanometer can be converted to an Ammeter by connecting a low resistance wire called a **Shunt** in **Parallel** with the Galvanometer
106. A Galvanometer can be converted to Voltmeter by connecting a high resistance or **Multiplier** in **Series** with the Galvanometer
107. The Potentiometer is a device used to precisely measure potential differences
108. The Wheatstone bridge is used to measure resistance
109. The meter bridge is used to measure resistance
110. Soft Iron is pure Iron
111. Steel is an alloy of Iron & Carbon
112. Steel is a much harder & stronger material than soft iron
113. Iron is more easily magnetized than steel, but it also loses its magnetism more quickly than steel
114. Iron produces a stronger magnet than steel
115. Steel is used for making permanent magnets such as compass needles
116. Iron is used for making temporary magnets such as electromagnets, where strong magnetism is required for only a short time. Hence, the core of an electromagnet is soft iron
117. A charge moving parallel to a magnetic field experiences no force
118. Lenz's law gives the direction of the induced current or emf
119. Lenz's law is an example of the conservation of energy principle
120. A Dynamo is a machine that converts mechanical energy into electrical energy or electrical energy into mechanical energy
121. When a Dynamo changes mechanical energy into electrical energy, The Dynamo is called a Generator
122. When a Dynamo changes electrical energy into mechanical energy, The Dynamo is called a Motor
123. D.C Generator: To convert an A.C to D.C, the two slip rings are replaced with a single split ring or commutator as in electric motor
124. The split rings/Commutator is a current reverser
125. An electroscope is an instrument used for the detection & testing of small electric charges
126. In the absence of air resistance (In a vacuum) or friction, all bodies fall with the same acceleration irrespective of their masses
127. A cell is a device for converting chemical energy into electrical energy
128. Local action is due to the impurities in zinc plate (Iron & carbon), this leads to the dissolution of the zinc plate into solution, even when no current is being supplied to the external circuit. The zinc is continuously wasted
129. Local action can be prevented by amalgamation. That is the rubbing of mercury over the surface of the zinc plate. The mercury on the zinc plate prevents the impurities fro coming into contact with the acid, with this, local action cannot occur
130. Dams in Northern Nigeria (Dam power station) → Kainji, Jebba, Shiroro
131. Rheostat is a variable resistor
132. NEPA → National Electric Power Authority
133. Argon is used in electric filament lamps, it prevents the hot filament from evaporating and enables the lamps to be run at high temperatures

134. Fuse rating is the maximum safe current permitted to flow in the fuse before it breaks

135. Surface tension is the force acting along the surface of a liquid, causing the liquid surface to behave like a stretched elastic skin

136. Surface tension is due to the forces of attraction between the molecules of the liquid (Cohesive forces)

137. Capillarity/capillary action is the tendency of a liquid to rise or fall in a narrow tube

138. Viscosity is the internal friction between layers of a liquid or gas in motion

139. When two plane mirrors are placed parallel to each other, the number of images formed is infinity

140. Concave mirrors are used as shaving mirrors

NOTE: A man should place his face between the principal focus & the pole of the mirror to get an enlarged, erect & virtual image of his face

141. Concave mirrors are used as reflectors in reflecting telescopes & microscopes

142. Convex mirrors are used as driving mirrors of cars because they give an erect image of an object behind the driver, they also provide a wide field of view

143. Some DISADVANTAGES of the convex mirror when used as a driving mirror

- The image is always smaller than the object
- It gives a false impression of the distance as the image seems farther away

144. Use of Parabolic mirrors in Car headlamps & search lights

- A **Parabolic mirror** is a special type of **concave mirror** which has the shape of **a parabola**
- The Parabolic mirror produces a **wide parallel beam of light of constant intensity** when a small light source is placed at its focus
- Parabolic mirrors are used in car headlamps & searchlights, Spherical mirrors are not used in these devices because they do not provide a **parallel beam of constant intensity**

145. Eclipse of the moon → SEM → The Earth is between the Sun & the Moon

146. Eclipse of the sun → SME → The Moon is between the Sun & the Earth

147. The image formed in a plane mirror is virtual, upright, same size as object, laterally inverted & same distance from the mirror as the object

148. A real image is one that can be caught on a screen

149. The critical angle is the angle of incidence in the denser medium when the angle of incidence in the less dense medium is 90°

150. Conditions for Total Internal reflection

- Light must travel from an optically more dense medium to an optically less dense medium
- The angle of incidence in the denser medium must be greater than the critical angle

151. Total Internal Reflection in Nature

- Mirages
- A fish can see everything in the air above the water within a cone of total angle 98° (half angle 49°)

152. Plane mirrors are not used in high quality periscopes because they produce multiple images due to multiple reflections from the mirror surface

153. Glass prisms with angles 90°, 45°, 45° are used in high quality periscopes as reflectors. These right angled Isosceles prisms have the advantage that only one image is produced

154. A right angled prism can also be used in a prism binocular to produce upright images that would otherwise appear inverted to the viewer

155. A prism binocular uses a combination of right angled isosceles prisms to achieve a high magnification without being as long as a telescope

156. Prisms have the advantage that they do not tarnish and deteriorate as mirrors do

157. Mirrors absorb more light and give a fainter image
158. The Dispersion of white light is due to the fact that the different colors of white light travel at different speeds through glass
159. Red light is deviated the least (Change of speed on entering the glass prism is least). Therefore, red light travels in glass with the greatest speed
160. Violet light is deviated the most, Violet light travels in glass with the least speed
161. In a vacuum however, all the colours travel with the same speed
162. The rainbow is formed by the dispersion of sunlight by spherical raindrops floating in the air
163. Monochromatic light is light of 1 wavelength or 1 colour
164. When monochromatic light is passed through a prism, refraction occurs without dispersion
165. To produce a pure spectrum we use 2 converging/convex lenses and a prism
166. To obtain a clear spectrum, we use a very fine slit & a bright light source
167. Primary colours → Red, Blue, Green
168. Secondary colours → Yellow, Cyan, Magenta
169. Biconvex & Biconcave lenses are common in a school laboratory
170. Plano-Convex & Plano-concave lenses are used in optical instruments
171. The eye has a crystalline biconvex lens (converging)
172. Polarization occurs in transverse waves (light) & not in longitudinal waves (sound)
173. Pitch and loudness have no effect on sound velocity
174. Pressure of surrounding air has no effect on sound velocity
175. Sensation of sound persists in the ear for about 0.1 seconds
176. The minimum distance away from the wall for a distinct echo is about 16.5m/17m
177. Reverberation is the persistence of sound after the source ceases
178. Reverberation is reduced in halls by covering the ceilings & walls with soft perforated boards which can minimize the reflection of sound by absorbing them quickly. Other methods are by hanging curtains round the hall & by having more openings in the wall
179. Loudness depends on Intensity & on the square of the amplitude of the sound wave
180. Quality depends on harmonics or overtones present in a note
181. Quality or tone distinguishes a characteristic note of an instrument from another note of the same pitch & loudness produced by another instrument
182. A musical note possesses the characteristics of pitch, loudness & quality